

Smartphone Battery Dynamics: Predicting Time-to-Empty

Summary

Our model aims to seek accurate Time-to-Empty predictions and effective power-saving strategies. As modern devices integrate diverse functions like computation, display, and network, each contributing to energy drain. Users also need clarity on which component dominates consumption.

For Task 1, we develop **SOC Time Evolution Model** to simulate the evolution of State of Charge and smartphone **temperature**. The model takes thermal-electric coupling into account. We start from basic energy conservation law and Newton's Cooling Law and establish Power Consumption Model and Temperature Model. Then, we decompose total power consumption into 3 parts: **CPU, Network, and Display**. We establish 3 submodel to describe these 3 modules respectively. Finally, we obtain SOC Time Evolution Model, which can describe different usage scenarios, network conditions, screen brightness settings and environment temperature. We determine the parameters in model by referring published data and conducting **CPU Power-Frequency Experiment** with **REDMI K80** smartphone.

For Task 2, we apply our model to predict Time-to-Empty under different conditions. We test Time-to-Empty in **4 different usage scenarios**: playing game (3.8hours), watching video (3.2hours), web browsing (6.7hours) and stand-by (25.7hours). We further conduct **uncertainty analysis** by introducing stochastic fluctuations in key parameters, including CPU frequency, download rate, and environment temperature. Then we simulate the effect of different **network condition, brightness setting** and **environment temperature** on Time-to-Empty.

In addition, we establish a **Battery Aging Model** based on **Arrhenius law** and compute the fractional capacity loss under different usage scenarios, revealing the long-term impact of operating conditions on battery health.

For Task 3, based on the simulation results and sensitivity analysis, we provide practical recommendations for smartphone users. We suggest that the most effective power-saving strategy is **changing the way of using smartphone**: reduce high-power consumption activities on smartphones, including playing games and watching videos. Besides, we find it inefficient to save energy by lowering brightness and improving network signal.

For Task 4, a sensitivity analysis regarding the linear coefficient k_0 is conducted, demonstrating that the model maintains stability with moderate sensitivity to parameter perturbations.

Keywords: lithium-ion battery; power consumption; battery drain; smartphone

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1 Introduction

1.1 Background

Smartphones have become indispensable tools in people's lives. As a highly integrated technological product, they fulfill our numerous demands for work, entertainment, communication, and more. The **Central Processing Unit (CPU)** receives and processes task instructions from various components of smartphone, performing mathematical calculations and data processing while coordinating the work of other parts through the controller. The smartphone **display** serves as the primary interface for user interaction. It conveys information to users by displaying various images, while users input information through touching the screen. **Wireless networks** are equally indispensable for smartphone operation. In the age of internet communication, people need to download diverse data and information—such as videos, spreadsheets, articles, and more—from the internet or various applications. During this process, the smartphone's network connectivity module must engage in bidirectional data exchange with base stations.

All of the aforementioned activities require the power from the smartphone battery to function. Therefore, when users evaluate the performance of a smartphone, the factor they often focus on is the battery's endurance. In specific usage scenarios, such as watching live video streaming, chatting on social media, playing games, etc., how fast the phone battery drains and how long the current battery level can sustain the normal operation of the phone are the issues that people are eager to know accurately during their use of the phone.

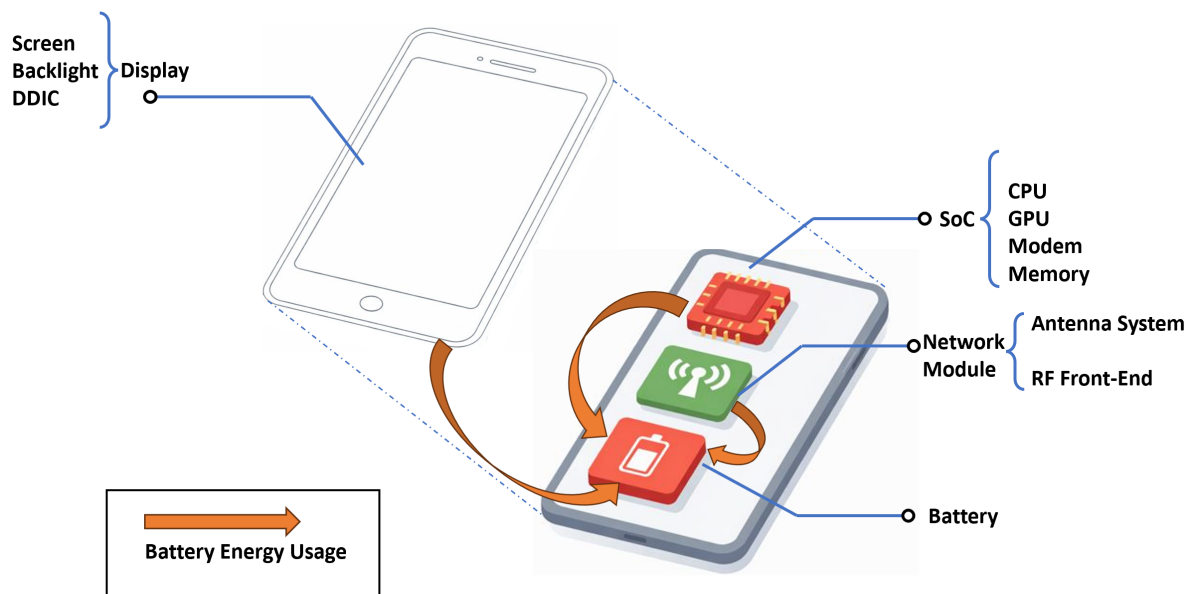


Figure 1: This picture illustrates the different functional modules of a smartphone that are primarily considered in this paper, each of which contributes to battery energy consumption.

By establishing time evolution models of battery **State of Charge (SoC)** in different usage environments, we can assist users and smartphone manufacturers in predicting battery **power consumption** and Time-to-Empty. Moreover, based on our model, we can recommend measures to reduce the rate of battery power drain. For mobile phone design manufacturers, we have proposed strategies to make the operating system more power-saving. Finally, we have constructed a model

for the aging of mobile phone batteries.

1.2 Restatement of the Problem

Based on the background mentioned above, our model should address the following tasks:

- **Task1:** Establish a Continuous Time-Evolution Model to describe how battery drain at different usage conditions, considering contributors such as CPU computation, network connection, display usage, background tasks and environmental temperature. Determine parameters in the model according to published measurement data.
- **Task2:** Use the model to estimate time-to-empty and compare the prediction in different scenario. Moreover, we should quantify uncertainty and evaluate the performance of our model.
- **Task3:** Make recommendations based on the model prediction. For smartphone users, illustrate what strategies can help save battery energy efficiently and how to slow down battery degradation. For smartphone manufacturer, propose how might an operating system implement more effective power-saving strategies.
- **Task4:** Examine how model predictions vary after making changes in modeling assumptions, parameter values, and fluctuations in usage patterns.

1.3 Our Work

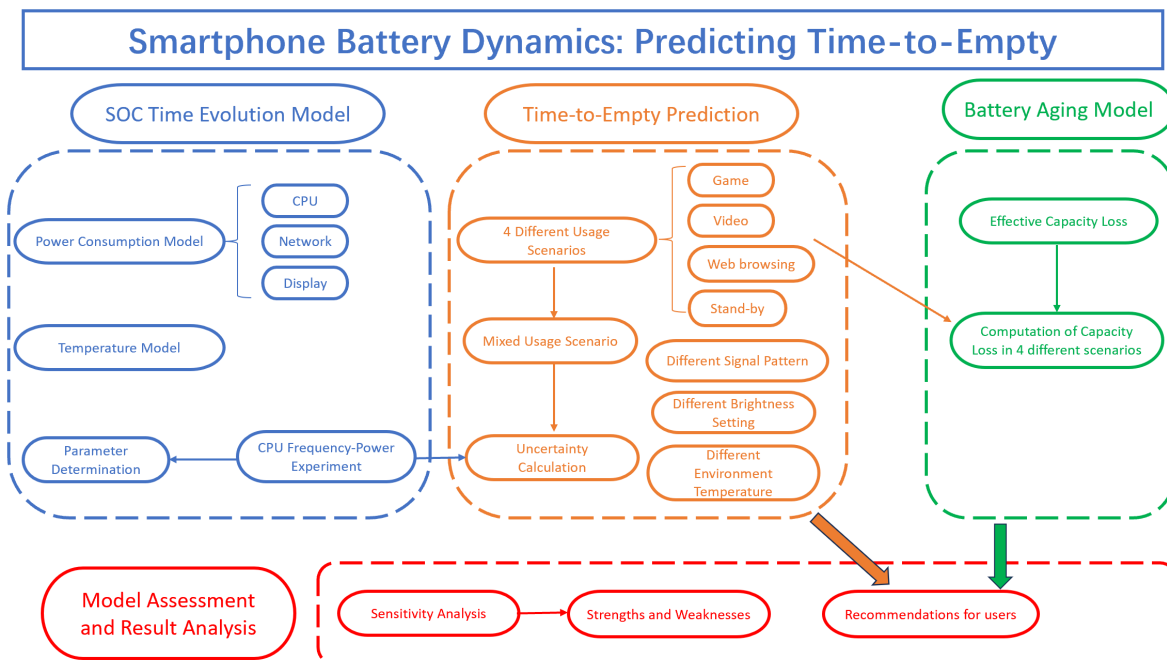


Figure 2: Our Work

2 Assumptions and Justifications

- **Assumption 1:** Except for possible battery aging effects, all smartphone components are assumed to remain in a healthy state and operate normally throughout the considered period.

→**Justification:** Under typical usage conditions, smartphone hardware components are designed to function reliably. Minor defects or performance fluctuations, if present, are assumed to have a negligible impact on the overall power consumption and are therefore ignored.

- **Assumption 2:** The smartphone is assumed to operate in a safe environment with a constant ambient temperature. Temperature is considered the only environmental factor affecting smartphone power consumption.

→**Justification:** Smartphones are generally well protected from environmental disturbances. Among external factors, temperature has the most significant and well-documented influence on battery performance and power consumption, making it reasonable to neglect other environmental effects.

- **Assumption 3:** The battery is assumed to be capable of supplying the required power demand of the smartphone until the state of charge (SOC) reaches zero.

→**Justification:** Modern lithium-ion batteries are designed with stable discharge characteristics over most of their operating range. Voltage drop and power delivery limitations near full depletion are neglected to simplify the continuous-time modeling of SOC evolution.

- **Assumption 4:** The primary sources of smartphone power consumption are network activity, CPU computation, and display usage. Power consumption from all other components is disregarded.

→**Justification:** According to recent studies in 2024 [1], across various usage scenarios, network communication, computational load, and display operation together account for over 80% of total smartphone power consumption. Thus, excluding secondary components introduces only limited error while significantly simplifying the model.

- **Assumption 5:** The power consumption from network connection primarily stemmed from downloading and processing data from internet.

→**Justification:** In most of usage scenarios in our life, smartphone download much more data than the data uploaded from the smartphone. Therefore, our model solely take power consumption when smartphone downloads information from network into consideration.

3 Notations

Symbol	Description	Unit
E_{eff}	effective battery capacity	W · h
t	time variable	h
SOC	state of charge	/
P_{tot}	total power of the smartphone	W
$P_{CPU}, P_{net}, P_{dis}$	power of CPU, network and display component	W
C	the effective heat capacity of the smartphone	J · K ⁻¹
η_0, η_n, η_d	the power efficiency of CPU, network and display component	/
f	CPU main frequency	GHz
f_0	default CPU main frequency	GHz
f_{dis}	screen refresh rate	Hz
$u(T)$	temperature correction factor	/
I_{down}	the average data download speed	MB/s
P_{max}	the maximum transmission power	MB/s
B	band width	Mbps
S	the area of the screen	m ²
ρ	pixel per area	m ⁻²
T, T_{env}	smartphone components temperature and environment temperature	°C
T_0	frequency reduction threshold temperature	°C
$RSRP$	downlink signal strength	dBm
SNR	signal noise ratio	dB
L	normalized luminosity	/

Table 1: Notation table

4 SOC Time Evolution Model

4.1 Model Overview

SOC Time Evolution Model describes battery drain speed based on the contributions of CPU, network and display. Then we establish CPU Model, Network Model and Display Model to evaluate their power consumption respectively. Temperature Model describes the impact of the heat generated by the operation of the above three modules on the smartphone's temperature. Moreover, the phone's temperature also affects the main frequency of the CPU and the refresh rate of display. The interrelationships of different parts in our model and the environmental input of our model are shown in Figure 2.

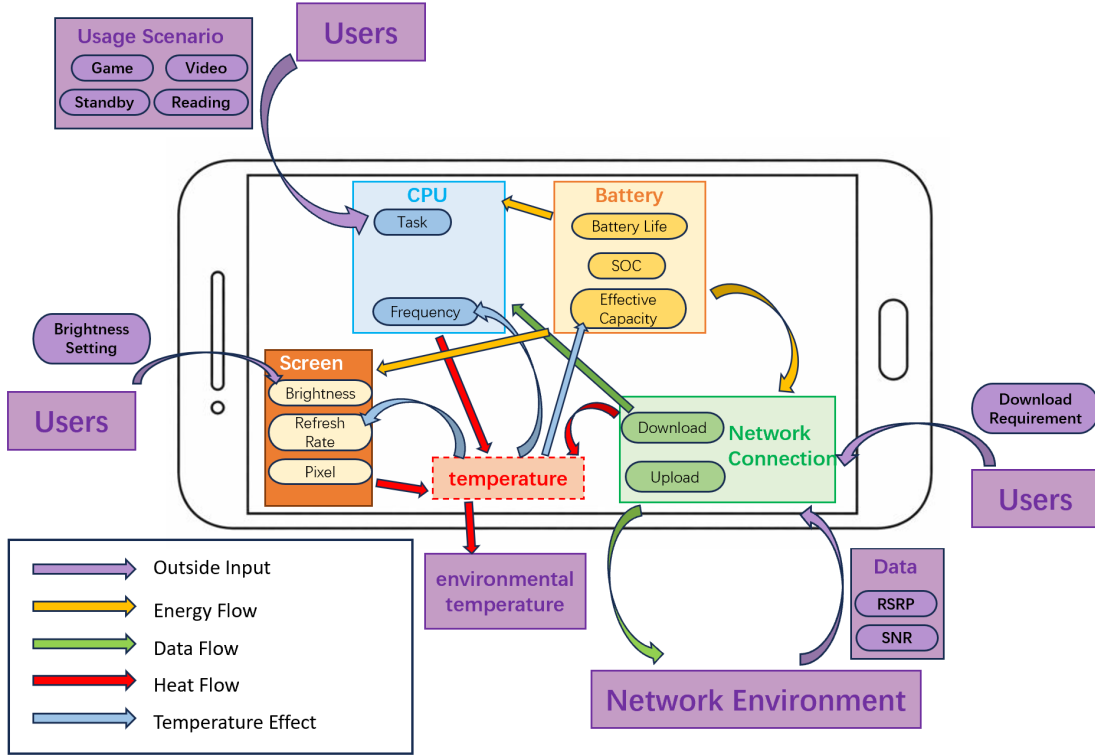


Figure 3: This diagram illustrates the interaction flow among different modules in our model.

4.2 Power Consumption Model

In a smartphone, POC refers to proportion of remaining battery energy. The time evolution of SOC depends on the power consumption from other modules of smartphone.

$$E_{eff} \frac{dSOC}{dt} = -P_{tot} \quad (1)$$

where E_{eff} refers to the effective total energy capacity of the lithium-ion battery, and P_{tot} refers to the total power consumption of the smartphone.

According to **Assumption 4**, We divide the total power into three parts: CPU, network and display (including screen and GPU).

$$P_{tot} = P_{CPU} + P_{net} + P_{dis} \quad (2)$$

It is worth noting that these power consumptions above includes both effective power and the wasted heat power. The heat generated in these processes increases the temperature of smartphone, which can be described by the following Temperature Model.

4.3 Temperature Model

The heat generated by the internal components of the mobile phone causes it to heat up. At the same time, according to Newton's cooling law, the phone also releases heat to the environment. The rate of heat release is related to the temperature difference between the smartphone's temperature

and the environment temperature.

$$C \frac{dT}{dt} = (1 - \eta_0)P_{CPU} + (1 - \eta_n)P_{net} + (1 - \eta_d)P_{dis} - \lambda S(T - T_0) \quad (3)$$

Where each η denotes the energy using efficiency of each components, defined as the proportion of effective energy in the total energy. The last term describes the transfer of heat to the environment. λ is a cooling constant determined by surface property.

High smartphone temperature will damage the hardware. To prevent the components from being damaged, the CPU and GPU generally apply throttling strategy. When the temperature is high, they reduce their operating frequency to produce less heat. In our model, we employ a temperature correction factor to represent the decrease in CPU main frequency as applying throttling strategy.[2]

$$f = f_0 u(T) \quad (4)$$

$$f_{dis} = 60u(T) \quad (5)$$

$$u(T) = \begin{cases} 1 & T \leq T_0 \\ u_{min} + (1 - u_{min})e^{-\frac{T-T_0}{\tau}} & T > T_0 \end{cases} \quad (6)$$

f is the average CPU main frequency and f_{dis} is the refresh rate of the screen. f_0 is the average normal frequency of a certain task. The default refresh rate of most phones is 60Hz. u refers to the temperature correction factor of CPU and GPU. T_0 refers to threshold of triggering throttling strategy

4.4 Effect of Temperature on Battery

Higher temperature can reduce the battery capacity. The battery will lose effective capacity under cold or hot conditions. We describe the effect using the following equation

$$E_{eff} = \frac{E_{eff}(25)}{F(25)} F(T) \quad (7)$$

$E_{eff}(25)$ is the effective battery capacity at 25 degrees celsius, considered a known parameter. From existing data [3], the temperature factor $F(T)$ is fitted as

$$F(T) = 0.0026T^3 - 1.2176T^2 + 73.8623T + 9933.3511 \quad (8)$$

We can see that at high or low battery temperature, the total effective capacity drops.

4.5 CPU Model

The CPU power consumption is determined by the software being used and the background processes. They are described by the following equation

$$P_{CPU} = \frac{1}{\eta_0} (P_0 + k_0 f + \alpha f I_{down}) \quad (9)$$

The first term comes from the background activities, which the CPU handle at a constant speed.

The second term comes from the current software in use or the major task of CPU. The frequency reflects the CPU processing speed. The higher the frequency, the greater the energy consumption.

The proportional coefficient k_0 is dependent on the current task. More demanding tasks like gaming will cause larger k_0 .

The third term describes the CPU power consumption required for processing the downloaded data. I_{down} is the average data downloading speed. α reflects the energy needed to process one unit of data.

4.6 Network Model

4.6.1 Working Mode

According to Assumption 5, we solely consider the power consumption from downloading data in network activities. When the smartphone downloads large amount of data frequently, it maintains a two-way signal transmission with the signal base station at all times in order to maintain a stable network connection. Therefore, the power consumption of the smartphone mainly comes from two parts: The Radio Frequency Front-End of the smartphone continuously sends signals to maintain two-way communication with the base station. Moreover, when receiving the signal, the smartphone needs to process and calculate the downloaded data. Therefore, network power consumption can be acquired as follow:

$$P_{net} = \frac{1}{\eta_n} (P_{RF} \frac{I_{down}}{v_{down}} + \beta I_{down}) \quad (10)$$

where

- P_{RF} refers to power consumption of RF Front-End.
- I_{down} refers to the average data download speed of the smartphone within a certain period of usage scenario.
- v_{down} refers to the download speed. $\frac{I_{down}}{v_{down}}$ is the proportion of time that Radio Frequency Front-End is active.

v_{down} and P_{RF} is determined by signal pattern of network, which is described by RSRP and SNR. RSRP refers to Reference Signal Received Power, SNR refers to Signal-to-Noise Ratio.

$$P_{RF} = \min(P_{max}, P_0 \cdot \frac{P_1}{10^{\frac{RSRP-30}{10}}}) \quad (11)$$

$$v_{down} = B \cdot \log_2(1 + \frac{SNR - 30}{10}) \quad (12)$$

where P_{max} limits the maximum transmission power P_{RF} smartphone using to keep connection to base station. P_0 denotes the minimum power requirement for the signals transmitted by smartphone to be effectively received by the base station when reaching the station. P_1 denotes the initial power of the download signal sent by the base station to the smartphone. Therefore, $\frac{P_1}{10^{\frac{RSRP-30}{10}}}$ expresses the signal loss rate during transmission.

4.6.2 Stand-by Mode

In stand-by mode, the smartphone downloads small amount of data because of the background apps. In this case, two important kinds of energy consumption become significant. The first one is ramping energy, which is the energy to activate the idle network receiving module to the receiving state. The second is the tail energy. At the end of data downloading, The receiving module will stay

in the high power receiving mode for a small period of time without receiving data.

$$P_{net} = \frac{1}{\eta_n}(f_n E_0 + \beta I_{down}) \quad (13)$$

f_n is the frequency of data downloading in the stand-by mode. E_0 is the sum of ramping energy and tail energy for a single download event. The second term is the same downloading energy as the active mode.

4.7 Display Model

The power consumption of display is:

$$P_{dis} = \frac{(k_1 L + k_2 f_{dis}) S \rho}{\eta_{dis}} \quad (14)$$

The first term in the formula describes the power consumption generated by the self-illuminating pixels (OLED) on the screen. It represents the most fundamental energy consumption of the display and is directly dependent on the screen brightness perceived by the user. The second term describes the power consumption of dynamic refresh and rendering, which represents the power consumption of the display driver circuit (DDIC) and the data flow of graphics processing.

4.8 Parameters Determination

In our model, based on the method for determining these parameters, there are 3 types of parameters:

- **Practical Parameters:** Parameters that can be determined based on hardware specifications, communication standards, or publicly available literature.
- **Scenario Parameters:** Parameters that changes based on different usage scenarios.
- **Fitted Parameters:** There are no direct measurements or reference value on these parameters. Therefore, they are fitted based on how long the battery will drain in practical and the power consumption proportion of 3 modules in our model.

For practical parameters and scenario parameters, we can determined them easily according to published measurements. These parameter settings are given in Table1.

Practical Parameter		Scenario Parameter	
Parameter	Value	Parameter	Value
$P_{\max}$ (dBm)	23	I_{down} (Mbps)	0.5
B (Mbps)	100	RSRP (dBm)	-80
f_0 (GHz)	2.0	SNR (dB)	25
R_0 (Hz)	60	L	0.3
S (m ²)	6.0×10^{-3}	$u_{\min}$	0.6
E_{base} (J)	8.72×10^4	T_0 (°C)	45
T_{env} (°C)	25	τ (°C)	5.0
ρ	0.7	E_0 (mJ)	150

Table 2: Related parameters

For fitted parameters given in Table 2, since these parameters are the linear coefficients in each of our sub-models, these parameters are not directly observable or independently measurable. They actually describe the complex characteristics related to the internal structure of the hardware and their values determine the proportion of power consumption among different modules of model. According to the experiment data [1] shown in Figure 4, we employ program to fit these parameters.

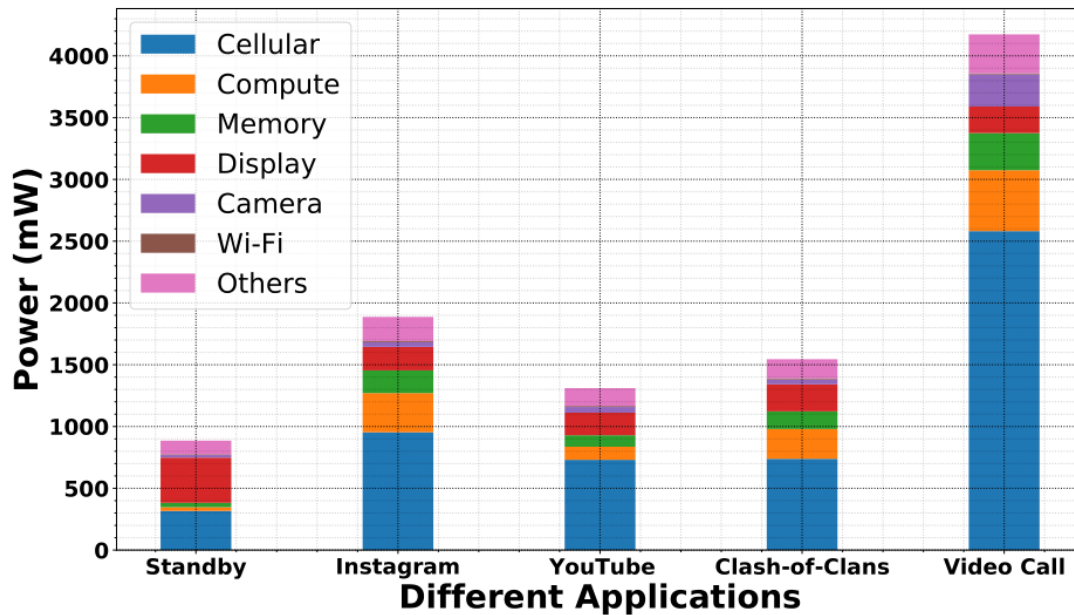
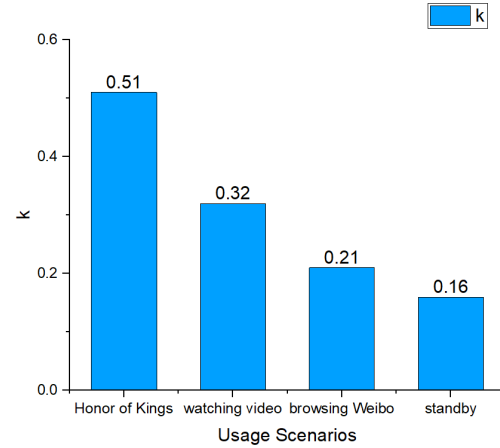


Figure 4: Reproduced from Ref. [1]. Average power consumption computed over 1-minute uses for different activities, like watching a 720p video on YouTube, doing a video call via Google Meet, scrolling on Instagram feed, playing Clash-of-Clans and keeping the phone idle in standby.

Parameter	Fitted Value
β	2.0
P_{CPU}	0.05
α	0.01
k_1	0.1025
k_2	0.0106

Table 3: Fitted Parameters

Figure 5: The fitting results of k_0 in four different usage scenarios.

In particular, for the coefficient k_0 , which significantly affects how CPU power consumption changes as CPU frequency varies, we conducted **CPU Power-Frequency Experiment** to measure the linear coefficient between the CPU power consumption of the smartphone and CPU main frequency. In the experiment, we tested the **REDMI K80** smartphone under four conditions: watching videos, playing Honor of Kings, browsing Weibo, and stand-by mode. The smartphone is connected to the computer, and a python program is used to collect data about CPU main frequency and power. The algorithm is shown below. Based on the experimental data of CPU power consumption and the CPU main frequency, we fit the value of k_0 and show the result in Figure 5.

Table 4: Algorithm of CPU Power-Frequency Experiment

Require: Sampling interval Δt , Refresh interval T_{pulse} , Capacity C_{rated}	
Initialize: CSV file, $t_{last} \leftarrow 0$, $I_{est} \leftarrow 200\text{mA}$, $Q_{real} \leftarrow \text{null}$	
While: collection is active do	
1. $t_{now} \leftarrow \text{TIME}()$	
<i>▷ Phase 1: Hybrid SOC Estimation (Pulse Strategy)</i>	
2. If: $(t_{now} - t_{last}) > T_{pulse}$	
• ADB(“dumpsys battery reset”)	▷ Wake hardware
• $Q_{real} \leftarrow \text{PARSECC}(\text{“charge counter”})$	▷ Read Coulomb Counter
• ADB(“dumpsys battery set usb 0”)	▷ Freeze charging
• Update I_{est} using $\Delta Q_{real}/T_{pulse}$	
• $t_{last} \leftarrow t_{now}$	
Else	
• $Q_{used} \leftarrow I_{est} \times (t_{now} - t_{last})$	▷ Dead Reckoning
• $Q_{current} \leftarrow Q_{real} - Q_{used}$	
End	
3. $SOC \leftarrow (Q_{current}/C_{rated}) \times 100$	
<i>▷ Phase 2: Peripheral Data Acquisition</i>	
4. $T_{batt}, T_{amb} \leftarrow \text{READSYSFS}(\text{“thermal_zone”})$	
5. $P_{cpu} \leftarrow \text{CALCPower}(V_{batt}, I_{est}, \text{Freq}_{cpu})$	
6. $Net \leftarrow \text{GETSIGNAL}(\text{RSRP}, \text{SINR})$	
7. $B_{scr} \leftarrow \text{GETBRIGHTNESS}()$	
8. Output: Write row $\{t_{now}, SOC, I_{est}, T_{batt}, P_{cpu}, Net, B_{scr}\}$	
9. $\text{SLEEP}(\Delta t)$	
End	
ADB(“dumpsys battery reset”)	▷ Restore system state

5 Time-to-Empty Prediction

In this section, we change scenario parameters, which can be controlled by users or environment and simulate Time-to-Empty by applying SOC Time Evolution Model. When one parameter varies, other parameters keep the same as Table2, Table3 shows.

5.1 Comparison among Different Usage Scenarios and Uncertainty Analysis

Scenario parameters k_0, I_{down} changes based on different usage scenarios, we select four typical usage scenarios(Game, Video, Web Browsing, Stand-by) and predict their Time-to-Empty. I_{down} is derived from the typical value in our life (Table6). k_0 is derived from the experiment mentioned in Parameters Determination subsection.

Scenario	k_0	I_{down}
Game	0.51	0.5
Video	0.32	1
Browsing Web	0.21	0.2
Stand-by	0.16	0.01

Table 5: Scenario Parameters Setting

Meanwhile, we conduct the uncertainty analysis of Time-to-Empty. To analyze it, we perform a Monte Carlo simulation by adding variance to three parameters: f_0, I_{down}, T_{env} , which are three significantly fluctuating parameters during the use of smartphone.

parameter	variance
f_0	0.4371
T_0	0.5
I_{down}	$\frac{1}{4}I_{down}$

Table 6: Variance of parameters

usage scenario	mean time-to-empty / h	standard deviation
game	4.1	1.2
video	3.3	1.1
word	6.6	1.7
stand-by	26.8	5.06

Table 7: Time-to-empty and standard deviation

The variance of f_0 is evaluated from the measured data and the other two is estimated based on practical scenarios. These three parameters vary in each round of calculation for 100 times and we obtain the Time-to-Empty and their uncertainty in each usage scenarios respectively (Figure6, Table 7). From the figure we can see that the uncertainty of Time-to-Empty is much larger than the other 3 scenarios because the time it takes for the smartphone battery to run out when in standby mode is much longer than in other situations, which leads to an increase in the variance of the final results.

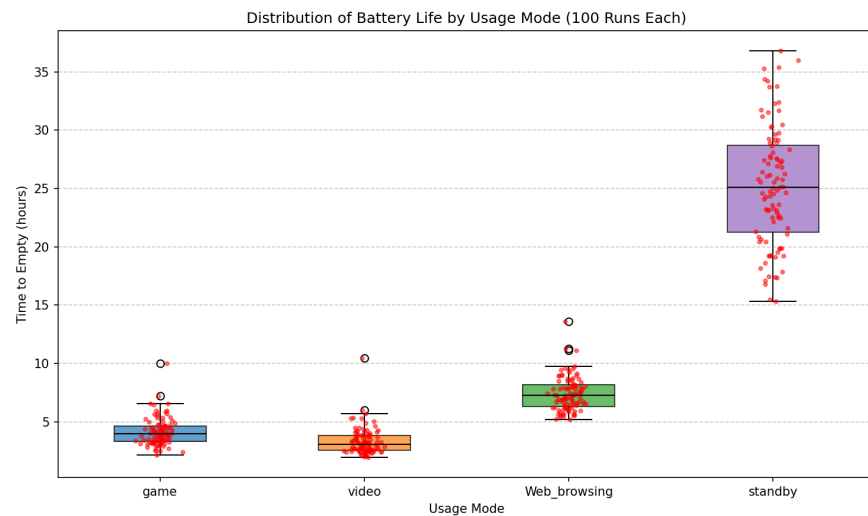


Figure 6: Box plot of Time-to-Empty prediction results under 4 usage scenarios after introducing noise

We can also identify the driver of battery drain in each scenarios, as Figure7 shows. We can conclude that:

- **Game Scenario:** The power consumption of the three modules is relatively close. Among them, **CPU** is the main driver of energy consumption, accounted for 40.6%.
- **Video Scenario:** **Network** contribute 55.5% of total power consumption. More data download requirement emerges when users watching videos thus network consumption dominates.
- **Browsing Web Scenario:** Browsing web impose little demand on downloading and computation and **display** becomes main driver, accounting for 45.9% of power consumption. The power consumption from merely browsing the web is about half that of playing games or watching videos.
- **Stan-by Scenario:** When smartphone is in stand-by mode, there is no display power consumption and CPU accounts for 93.8% power consumption because CPU still need to process **background tasks**.

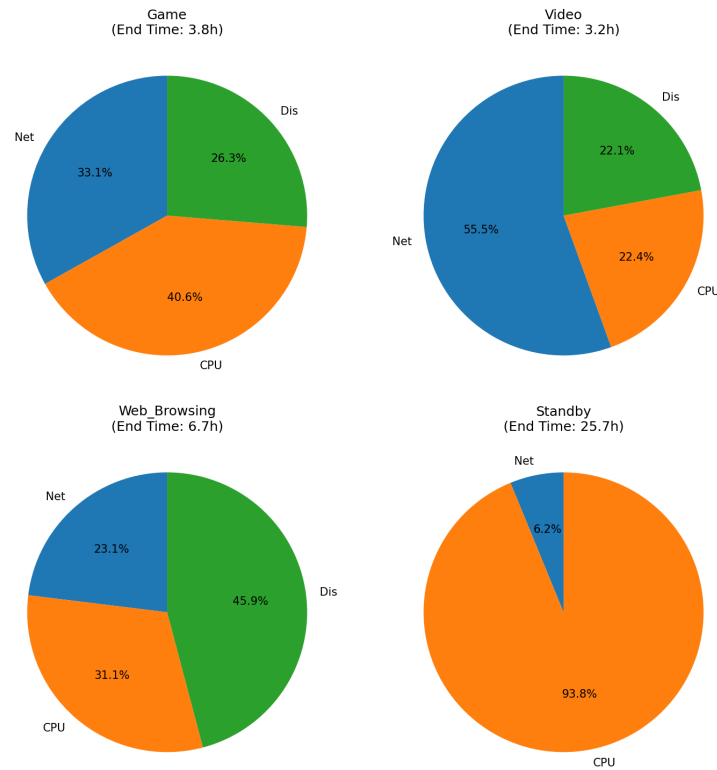


Figure 7: Pie chart showing the energy consumption proportions of 3 smartphone modules in 4 usage scenarios

5.2 Mixed Usage Scenario

In the aforementioned simulation, in order to control the variables, we calculated the SOC time evolution under a single usage scenario while keeping other variables constant. We also calculated the Time-to-Empty. In actual situations, the user's usage behavior throughout the day can be a combination of the aforementioned usage scenarios. Moreover, the CPU frequency, environment temperature, and download data speed often fluctuate over time. To simulate the real usage scenarios in our life, we used a program to randomly generate a set of input scenarios representing different combinations of usage scenarios, and added small-range Gaussian noise to the parameters to simulate the real fluctuations. The possibility of four scenarios are the same. Each section of the scenario lasts 20-40 minutes.

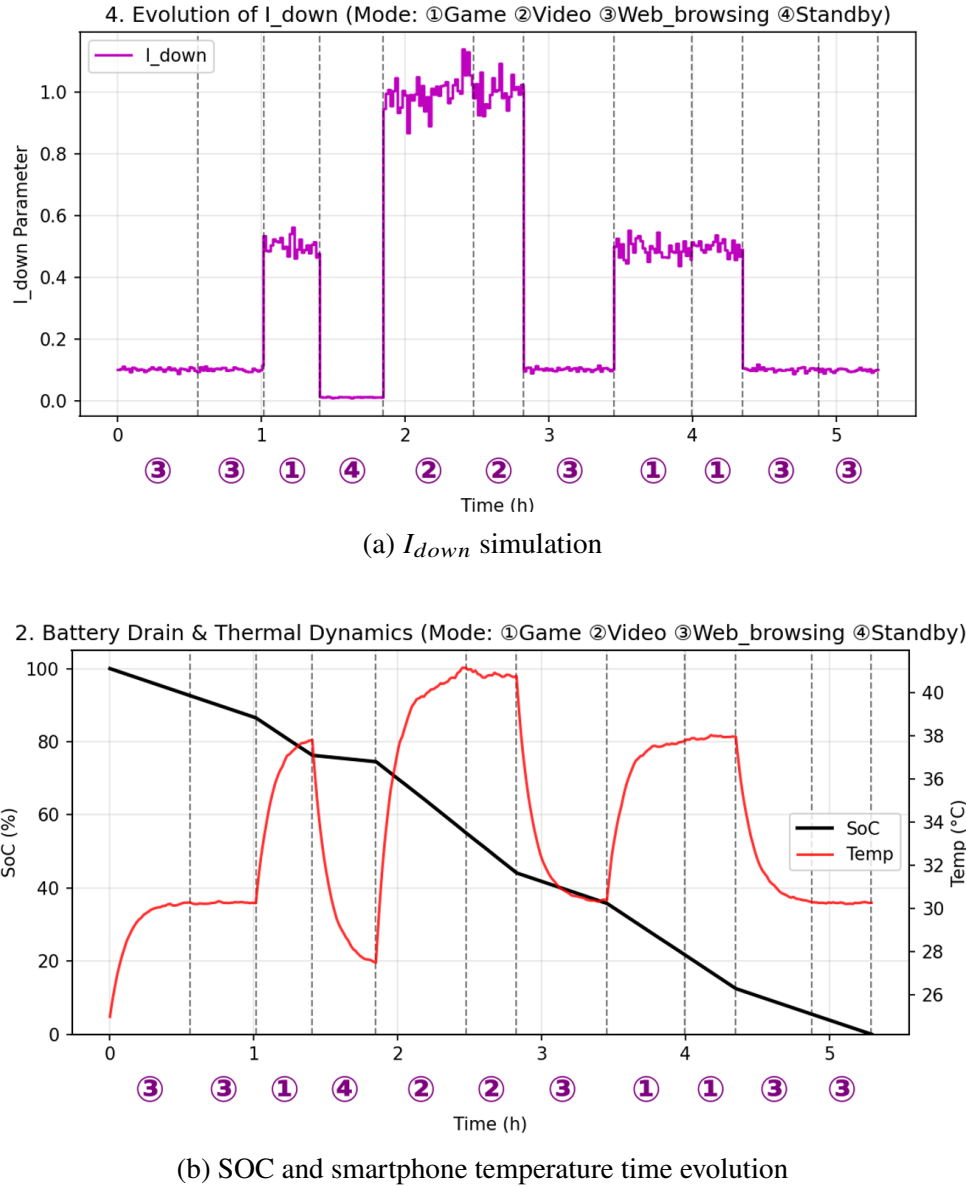


Figure 8: Practical Scenario SOC Time Evolution Simulation

This evolution result is reasonable since it explicitly illustrates the speed of battery drain changes as usage scenario switches and the temperature fluctuates continuously under different usage modes.

For arbitrary usage scenario, we can input it into the time evolution equation and predict Time-to-Empty.

5.3 Effect of Brightness of Display

According to SOC time evolution model, time-to-empty can be simulated with different brightness settings. We changes brightness from 1 to 0 and the simulation result is shown as follows:

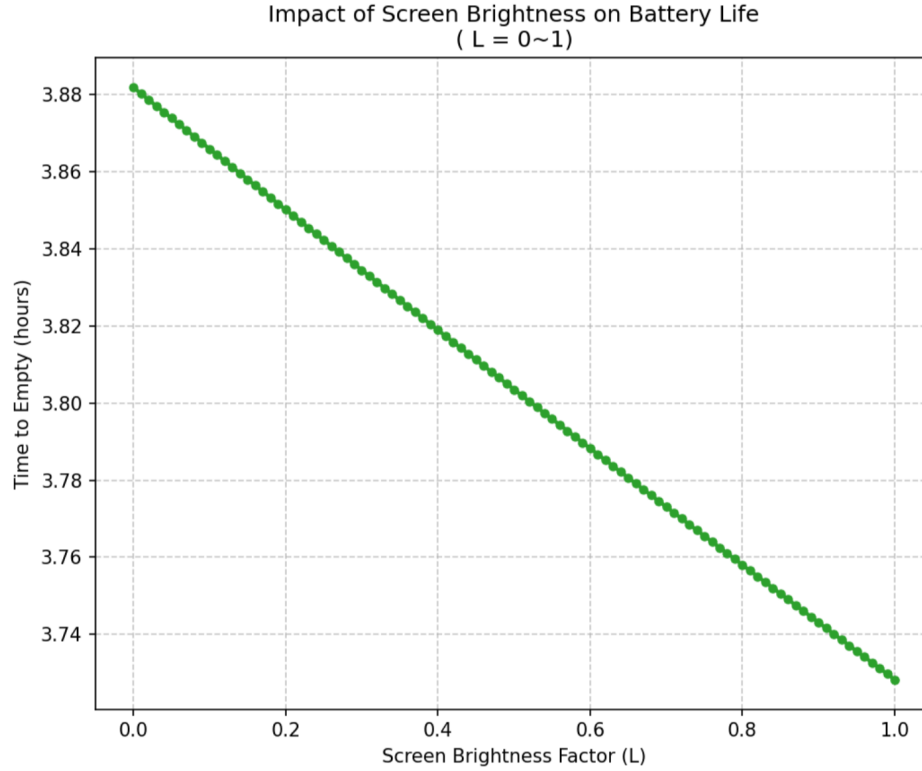


Figure 9: Time-to-Empty with different brightness settings.

From Figure 9, we can see that if the screen brightness is adjusted from the brightest to the dimmest, the Time-to-Empty of the phone will decrease by approximately 4%.

5.4 Effect of Signal Condition

In practical usage, the signal environment of smartphone varies under different conditions. For instance, in WiFi and cellular network modes, SNR and RSRP will change due to the difference in the distance of the base station. In our simulation, we vary the SNR value within the range of $[-30, 30]$, and the RSRP value vary within the range of $[-120, -30]$. The simulation results are as follows:

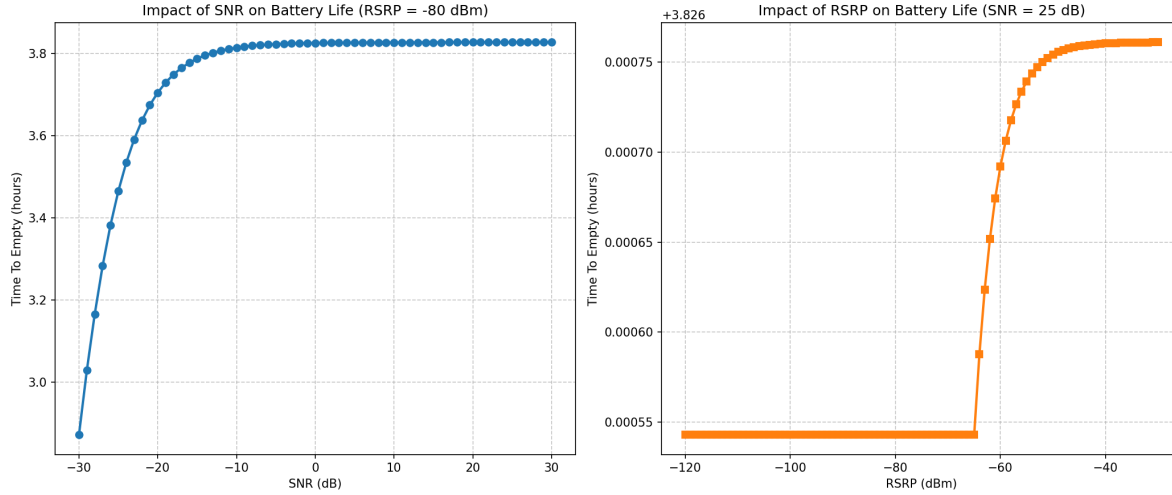


Figure 10: Time-to-Empty with different signal environment.

It can be seen that RSRP has almost no impact on Time-to-Empty, this is because, in our model, P_{max} determines the maximum value of the transmission power $P_R F$. Therefore, the smartphone will not increase its upload transmission power without limit in the case of poor signal RSRP. Therefore, according to our simulation, the impact of RSRP on the power consumption of the smartphone can be ignored.

Moreover, as SNR increases from -30 dB to about -10 dB, Time-to-Empty increases rapidly. Beyond this range, further improvements in SNR lead to only marginal gains, and the curve gradually saturates. When SNR is low, the communication link operates in a highly inefficient regime: more transmission power and longer active time are required to maintain data throughput, leading to substantial power consumption in the network module. When network connection is well, which means SNR always stay more than 10, the improvement of SNR can not improve Time-to-Empty significantly.

5.5 Effect of Environment Temperature

Impact of environmental temperature is demonstrated in Figure11. In our setting, when the temperature of smartphone increases over $T_0 = 45^\circ\text{C}$, throttling strategy will be triggered, CPU will reduce its main frequency and GPU will reduce refresh rate of display in order to reduce the heat production. This mechanism is illustrated in the figure, when environment temperature reaches 32°C , the stable value of the smartphone temperature at thermal equilibrium will reach over 45°C , CPU frequency and screen refresh rate decreases. The descend of power consumption of CPU and display significantly increases Time-to-Empty. However, it is worth noting that although Time-to-Empty is extended, the user experience can deteriorate due to the decrease in CPU frequency and screen refresh rate. At this point, the display smoothness and the computing speed will both decline, and the phone will become sluggish.

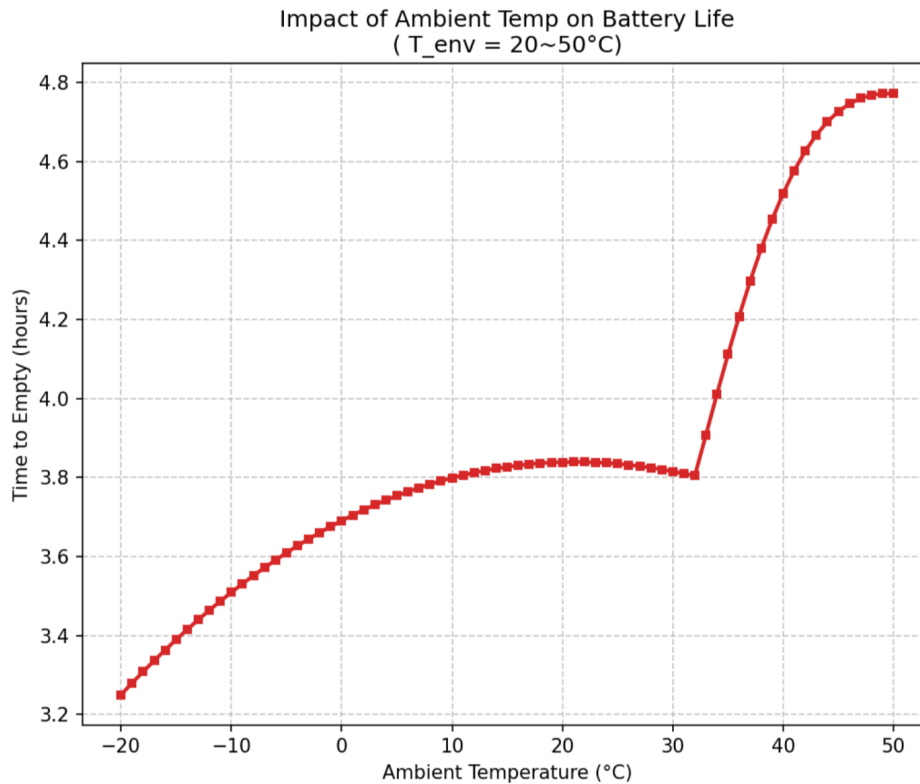


Figure 11: Time-to-Empty with different environment temperature.

5.6 Effect of Background Load

The background load is reflected in the CPU static power P_0 . When the smartphone is actively working, the background power only accounts for a little proportion of total power. During the stand-by mode, however, the static power becomes significant.

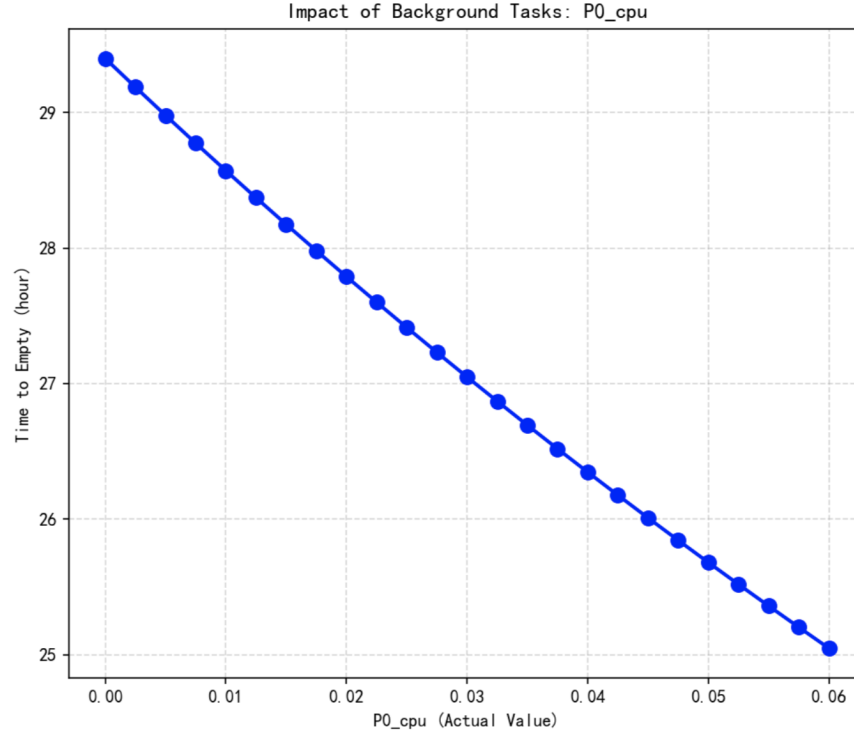


Figure 12: Time-to-empty prediction with different background power

We can see that during the stand-by mode, a 0.01W change in the CPU background power will lead to approximately 40 minutes more battery usage time, which is a large improvement.

5.7 Conclusion

According to our simulation, for short, we can conclude that the impact of different environment or user setting on Time-to-Empty:

Environment Temperature > Brightness » Network Environment

Time-to-Empty in different usage scenarios,

Play games ≈ Watching video < Browsing Website « Stand-by

6 Battery Aging Model

Temperature is one of the most significant factors in battery aging. We describe the aging as the gradual drop in effective battery capacity. The capacity drop is because of several chemical reactions inside the battery that lead to loss of lithium, active material and conductivity.

$$\frac{dE_{eff}}{dt} = -kE_{eff} \quad (15)$$

k is determined by the Arrhenius law [4]

$$k = A \cdot \exp\left(\frac{E_a}{RT}\right) \quad (16)$$

A is a constant pre-exponential factor, R is the universal gas constant and E_a is the activation energy.

In a single discharge cycle, the change of E_{eff} is small, so we can take it as a constant at the RHS of equation 15.

We then use the $T - t$ relation given by the temperature model to calculate the total loss ratio of effective battery capacity in one discharge section.

$$\frac{\Delta E_{eff}}{E_{eff}} = 1 - e^{-\int_0^T k dt} \quad (17)$$

The empirical values of parameters are [5]

$$A = 10^6 \text{year}^{-1}, E_a = 50 \text{kJ} \cdot \text{mol}^{-1}$$

Using these parameters, we calculate the change ratio of battery

Scenario	Duration /h	Capacity loss ratio	Average temperature	Capacity loss ratio per hour
Game	3.82	1.73×10^{-6}	38.07	4.53×10^{-7}
Video	3.20	1.70×10^{-6}	40.78	5.31×10^{-7}
Web Browsing	6.66	1.99×10^{-6}	31.28	2.99×10^{-7}
Stand-by	25.68	6.00×10^{-6}	27.43	2.33×10^{-7}

Table 8: Total loss ratio in different scenarios

The stand-by scenario causes the most capacity loss due to its long duration. We can see that temperature has a significant impact on the reduction of battery capacity.

7 Recommendations for Users

Based on our findings we can make following recommendations for smartphone users on power saving and battery life extension: **What to do:**

- **Changing the way of using smartphone:** In certain situations, such as when no charging device is available, smartphone users may wish to extend the usage time. We suggest that the **most efficient way** for users to achieve this is to reduce high-power consumption activities on their phones, including playing games and watching videos. Instead, they should mainly use low-energy modes such as browsing text-based web pages and being in standby mode. This usually doubles the time before the battery runs out, although users may have to sacrifice some of their demands for the entertainment functions of the phone.
- **Reducing the screen brightness improves little:** Even if the user sets the phone display brightness to be very low, it can only extend the phone's usage time by at most 4%, but it significantly deteriorates the user's experience.

- **High temperature environment can negatively affect your mobile phone usage experience:** This is because in high-temperature conditions, the CPU will actively reduce its main frequency while the GPU will lower the display refresh rate to reduce heat generation. This will cause your phone's computing speed to slow down and the screen to lag. Higher temperature will also lead to faster battery aging. You should avoid using your phone in a high-temperature environment for a long time, such as using it under direct sunlight or running high-load tasks while charging. Additionally, you should actively cool down your smartphone to enhance its performance.
- **Reducing background processes can extend your stand-by time:** The CPU power consumption in standby mode accounts for as high as 93.8%, indicating that background tasks have a significant impact on battery life. We recommend that users regularly clear background applications, especially social applications (such as WeChat, Instagram) that remain in the background and frequently wake up the CPU. This can extend your standby time by approximately 2-4 hours.

8 Sensitivity Analysis and Stability Analysis

When establishing our time evolution model, we determine some parameters through experiment data and fitting results in the literature. In this Section, we conduct sensitivity analysis for the parameters k_0 , which we derive from the CPU power-frequency experiment mentioned in Parameters Determination subsection, since k_0 is an important parameter affecting the power consumption of CPU. Therefore, it is necessary to analyze model's sensitivity to k_0 because our experiment may have relatively large uncertainty. The result is shown in Figure 13, 14.

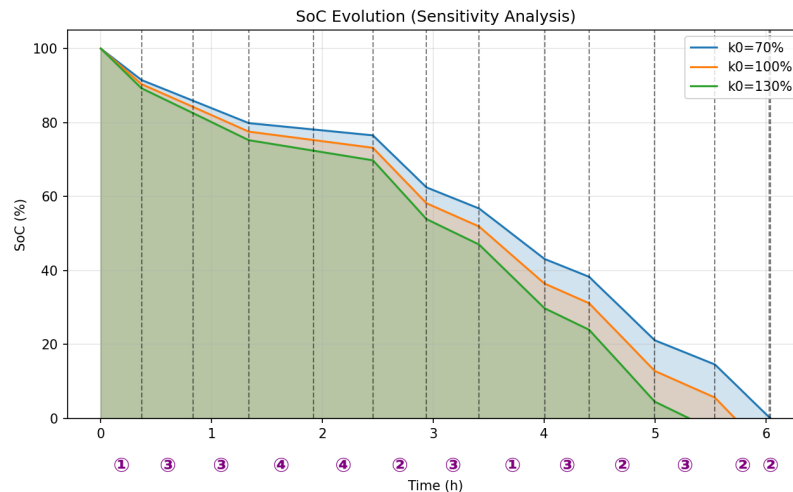


Figure 13: Time-to-empty evolution under different k_0

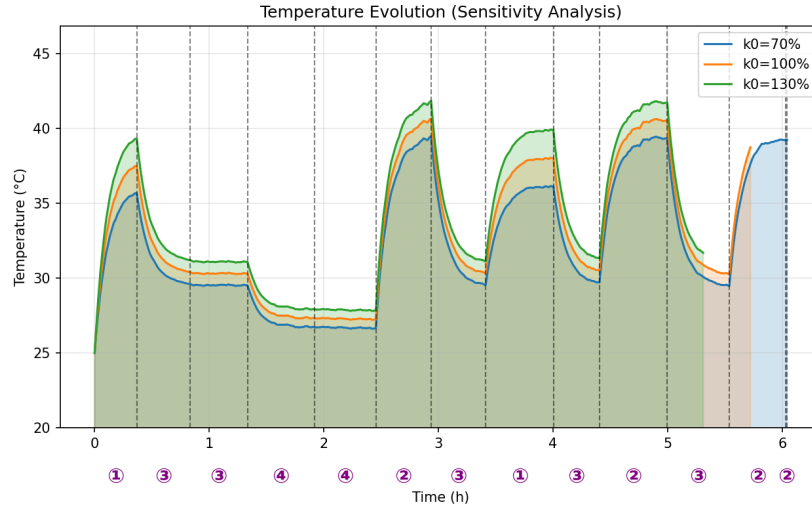


Figure 14: Temperature evolution under different k_0

It can be seen from the figure that our model shows moderate sensitivity to k_0 . Such a result is reasonable because the change in k_0 directly affects the impact of CPU frequency on power consumption. Moreover, the results of the sensitivity analysis indicate that the change in this parameter does not affect the SOC and temperature evolution trend of the model.

9 Strengths and Weaknesses

9.1 Strengths

- Our model is built upon a continuous-time SOC and temperature evolution equation directly derived from energy conservation, where battery depletion is governed by physically interpretable power consumption terms.
- For the CPU, Network, and Connection modules, we have constructed detailed sub-models. Each module can be independently explained, parameterized, and analyzed. This ensures that we can perform highly flexible model fitting based on actual usage data to enable the model to simulate real scenarios.
- The model is flexible and can be adapted to different environment conditions or user scenarios, such as high environment temperature and playing games, which allows for comprehensive application in different cases.

9.2 Weaknesses

- The fitted parameters are based on publicly available experimental data and a small amount of practical measurements by our team itself. This can make our prediction inaccurate compared to the results in our practical usage.
- Our model is insensitive to the signal environment. According to experimental data, it can actually lead to fluctuation of the battery drain speed by approximately 5%.

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- [2] Ravishankar Rao and Sarma Vrudhula. Performance optimal processor throttling under thermal constraints. In *Proceedings of the 2007 International Conference on Compilers, Architecture, and Synthesis for Embedded Systems*, CASES '07, page 257–266, New York, NY, USA, 2007. Association for Computing Machinery.
- [3] Shuaishuai Lv, Xingxing Wang, Wenfan Lu, Jiaqiao Zhang, and Hongjun Ni. The influence of temperature on the capacity of lithium ion batteries with different anodes. *Energies*, 15(1), 2022.
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- [5] Johannes Schmalstieg, Stefan Käbitz, Madeleine Ecker, and Dirk Uwe Sauer. A holistic aging model for li(nimnco)o2 based 18650 lithium-ion batteries. *Journal of Power Sources*, 257:325–334, 2014.

Report on Use of AI Tools

Trae-3.5.28

Query1:

In the file “Simulating Real Continuous Use of Phone.py” in the workspace, this models the frequent switching usage patterns of real users by introducing Gaussian noise to three parameters (with a total of four usage modes). The visualization code for Figures 1 and 2 is removed. Now, in this simulated environment, perform sensitivity analysis for k_0 .

The original code has four modes, each corresponding to a specific k_0 . The simulation runs once for each mode.

Modify it so that for each program run, the simulation runs three times. These three runs should correspond to the k_0 values of “70%“, “100%“, and “130%“ for the four modes.

For the "70%" group:

- First, generate Figures 3 and 4.
- Additionally, generate two more plots showing the changes in *SOC* (State of Charge) and temperature over time (similar to splitting the original Figure 2).
- Note that these two new plots must still show the corresponding mode category for each time point (as in the original method).

After fixing the "f0", "T_env", and "I_down" values over time, as well as the mode duration and type over time (as shown in Figures 3 and 4 for the "70%" group), run the simulations for the "100%" and "130%" groups.

- For the "100%" and "130%" groups, do not generate any additional plots.
- Instead, add the *SOC* and temperature data (over time) from the "100%" and "130%" simulations to the previously generated two plots.

The "70%" group acts as the "pioneer", where the battery is drained to 0, and a unique "path" is established. The "100%" and "130%" groups follow the "70%" path, but they are not as efficient. They will run out of power before the "70%" group.

The ideal result is to generate four plots:

1. **Figures 3 and 4**, which can already be generated.
2. Two additional plots, each containing three lines. These lines should all start at the same point but end at different times.
 - Ideally, color the area under the curve with the same color as the line, similar to the example provided, to make it easier to understand.
 - Ensure that in the additional plots, the colors for the same groups are consistent.

There are many requirements and the understanding is complex. Please take your time to carefully modify the code and thoroughly check everything to ensure that the four plots meet the ideal outcome. There's no rush.

Output1:

I have modified the code as per your requirements to perform the sensitivity analysis for the k_0 parameter.

The main changes are as follows:

1. Reconstruction of Simulation Logic:

- The simulation is divided into two stages:
 - **First Stage ("Pioneer" - 70% group):** This stage is responsible for generating random usage patterns (game/video/document/standby) and environmental noise parameters (f0, T_env, I_down, etc.) changing every minute. These parameters are recorded to form a fixed "path". The physical simulation is run with $k_0 = 70\%$.
 - **Second Stage ("Follower" - 100% and 130% groups):** This stage completely replays the environmental parameter path generated in the first stage, but applies the $k_0 = 100\%$ and $k_0 = 130\%$ parameters to run the physical simulation.
- This ensures that the three sets of experiments face exactly the same external conditions and usage patterns, with only the k_0 parameter differing.

2. Visualization Adjustments:

- The original Figure 1 (Power Stacking Graph) and Figure 2 (SOC and Temperature Mixed Graph) have been removed.

- Figures 3 (f0 and T_env) and 4 (I_down) are retained, both based on the 70% group ("Pioneer") data. These reflect the fixed environmental path for the current experiment.
- Two new plots have been added:
 - SOC over time: This plot includes three curves (70%, 100%, 130%) with color filling.
 - Temperature over time: This plot includes three curves (70%, 100%, 130%) with color filling.
- All four plots retain the mode category labels (1,2,3,4) at the bottom and dashed line separation.

You can directly run the modified script "Simulating Real Continuous Use of Phone.py" to view the results.

ChatGPT-5.2

Query1: What are the estimated percentage of energy generated by heat in each components of the smartphone? Give some authorized data source.

Output1: The energy generated by heat in smartphone components can vary depending on factors like device model, usage patterns, and hardware. However, based on studies of energy consumption in smartphones, here's a rough breakdown of where heat generation predominantly occurs:

- **Processor (CPU, GPU):**
 - **Estimated energy contribution:** 40-50% of the total heat generated.
 - **Reason:** The CPU and GPU are the most power-hungry components, particularly during resource-intensive tasks like gaming, video rendering, or multitasking.
- **Display:**
 - **Estimated energy contribution:** 20-30%.
 - **Reason:** The screen, especially high-resolution or OLED screens, consumes a lot of power, contributing to heat generation.
- **Battery:**
 - **Estimated energy contribution:** 10-20%.
 - **Reason:** While the battery itself does not directly generate heat, the power conversion and charging process can lead to heat, especially during fast charging.
- **Wireless communication modules (Wi-Fi, Bluetooth, Cellular):**
 - **Estimated energy contribution:** 10-15%.
 - **Reason:** Components responsible for data transmission (Wi-Fi, LTE, Bluetooth) also consume significant power, contributing to heat generation during active use.
- **Other components (storage, sensors, etc.):**
 - **Estimated energy contribution:** 5-10%.

- **Reason:** Components like the storage drive and sensors (e.g., accelerometer, gyroscope) use relatively less power but still contribute to overall heat.

Authorized Sources:

- “*Energy Efficiency in Mobile Devices*” by Open Mobile Alliance (OMA): This report delves into energy consumption in mobile devices and provides insights into the proportion of energy consumed by various components.
- “*Smartphone Power Consumption Models*” by M. Zhang et al. (IEEE Access): This research explores the power consumption models of smartphones and the energy breakdown across various components.
- “*Power Consumption in Mobile Computing Devices*” by R. Li et al. (IEEE Transactions): A detailed study focusing on power usage patterns in mobile devices, helping understand heat generation and energy distribution.